

Omar Khamis Abdelhafiez Barakat

+20 102 730 4125 | omar.khamis.barakat@gmail.com | omarkhamisbarakat.me | [GitHub](#) | [LinkedIn](#) | Cairo, Egypt

EDUCATION

Zewail City of Science and Technology

B.Sc. Communication and Information Engineering — ABET Accredited

Expected 2027

Cairo, Egypt

Relevant Coursework: Embedded Systems, Machine Learning, Statistical Inference & Data Analysis, Data Structures & Algorithms, Digital Design & Computer Architecture, Database Management, Information Security

EXPERIENCE

Deutschland Technologies

Engineering Intern

Jun 2025 – Aug 2025

Cairo, Egypt

- Designed fire-alarm systems integrating sensors, control panels, and output devices for commercial installations; documented wiring schematics using AutoCAD.
- Assisted in project planning and milestone scheduling using Oracle Primavera, supporting on-time delivery of safety-system rollouts.

DataAnnotation Tech

Data Annotator (Contract)

May 2025 – Feb 2026

Remote

- Annotated NLP datasets and delivered contextual feedback within RLHF pipelines used to train and align large language models.
- Contributed to human-in-the-loop evaluation workflows, improving model accuracy and response quality across diverse task domains.

PROJECTS

Autonomous Safety Car — Cyber-Physical Embedded System

PIC16F877A · Raspberry Pi · TensorFlow Lite · Flask · Socket.IO · Firebase · Docker · SQLite · Python · C

Spring 2026

- Engineered a layered cyber-physical vehicle: PIC16F877A owns deterministic motor control and safety enforcement; Raspberry Pi bridges web UI, cloud logging, and ML inference via UART at 9 600 baud.
- Implemented 5 firmware safety features — directional ultrasonic obstacle guard (25 cm threshold), active-stop detection, seat-belt interlock, PWM speed guard (warn at 80 % / stop-reset at 90 %), and LDR low-light LED assist — all enforced in firmware so the browser UI cannot bypass them.
- Trained and deployed a TC-ResNet-Lite TFLite keyword-spotting model (53 K params, 12 450 examples) with custom NumPy-only MFCC preprocessing on resource-constrained hardware; per-command confidence thresholds cut false-accept motion events.
- Architected dual-layer persistence (SQLite local + Firebase cloud mirror) with live Socket.IO dashboard; containerised full stack via Docker/Compose with systemd auto-start and self-signed TLS for browser microphone access.

Plant Disease Detection — Computer Vision

Python · TensorFlow / Keras · OpenCV · CNN

2025 – 2026

- Built a convolutional neural network classifier to identify disease categories from plant-leaf images, supporting precision-agriculture diagnostics.
- Implemented image-preprocessing pipeline (resizing, normalization, augmentation) and evaluated model performance across disease classes with confusion-matrix analysis.

ESP32-Based Digital Synthesizer — Embedded Audio

ESP32 · C++ · DSP · I2S / DAC

2025 – 2026

- Designed a real-time digital synthesizer on an ESP32 microcontroller, implementing DSP algorithms for multi-waveform audio generation with hardware user-input controls.
- Optimised signal-processing routines under strict real-time and memory constraints inherent to bare-metal embedded targets.

MLE & Estimator Distribution in Linear Regression — Statistical Analysis

Python · NumPy · Matplotlib · OLS · WLS · Monte Carlo

Spring 2026

- Derived OLS and WLS as MLEs under homoscedastic and heteroscedastic Gaussian noise; validated exact sampling distributions against 10 000-trial Monte Carlo simulations confirming WLS achieves $\sim 2\times$ lower slope variance than OLS under misspecified noise.
- Constructed confidence and prediction intervals from estimator distributions; proved estimator optimality via the Cramér–Rao lower bound on a real wage–age dataset.

TECHNICAL SKILLS

Languages: C, C++, Python, C#, MATLAB, Dart

Embedded & Hardware: PIC16F877A, Raspberry Pi, ESP32, UART / PWM / GPIO / I2S, MPLAB X / XC8, LabVIEW, AutoCAD

Machine Learning: TensorFlow / Keras, TFLite, CNNs, MFCC, Monte Carlo Simulation, MLE, OLS / WLS, OpenCV

Cloud & DevOps: Firebase Realtime DB, Docker / Compose, systemd, SQLite, Git

Software & Frameworks: Flask, Socket.IO, ASP.NET, Flutter, Bootstrap, HTML, Oracle Primavera, CST Studio